

AA411

D.A.P. System Test Plan: Phase B

D.A.P. System Team

Part Name: D.A.P. System

Part Number: DAPFS1

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Test Team:

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Introduction

The purpose of this system-level test campaign is to verify that the Damage Assessment Protocol (D.A.P.) system satisfies the system-level requirements related to onboard-data usage, fault-detection accuracy, and computational performance (SR-1, SR-2, and SR-4). The D.A.P. system integrates onboard Seaglider mission data, a calibrated Digital Twin, and the Diagnoser function to determine whether a vehicle is operating nominally or experiencing a single wing-loss or rudder-loss physical fault.

Although the original test plan included a Phase A field deployment to collect 15 new dives across nominal and faulted configurations, technical issues with the Seaglider prevented the acquisition of usable mission data. As a result, the system-level evaluation is conducted entirely using Phase B post-processing, based on a curated Test Matrix of previously recorded Seaglider mission files. These datasets, sourced from the Colvos, Edmonds, and Motive missions, provide a diverse set of operating conditions suitable for evaluating the D.A.P. system.

This test campaign verifies SR-1 by demonstrating that the D.A.P. system requires only the onboard-generated .log and .nc files to perform its analysis. The Test Matrix provides the filenames for each dive, and these filenames are the sole inputs needed to retrieve the mission data. No external measurements, metadata, pilot annotations, or off-board information are used at any point in the D.A.P. pipeline. All Digital Twin predictions, Diagnoser metrics, and final classifications are produced exclusively from the onboard mission files, confirming compliance with SR-1.

The campaign verifies SR-2 by evaluating the system's ability to detect single-wing-loss and rudder-loss faults using the mission datasets defined in the Test Matrix. During Phase B, the Digital Twin generates predicted vehicle behavior for each dive, and the Diagnoser compares these predictions to the recorded .nc files. Measurements used to validate SR-2 include percent-difference arrays, window scores, excursion intervals, and final fault classifications. A requirement pass is achieved when the system identifies the correct fault type within five data transmissions.

The campaign verifies SR-4 by measuring the total computational runtime required to process each dive dataset. The full D.A.P. pipeline is executed across all Test Matrix cases, and wall-clock runtime is recorded to confirm that each dive can be analyzed in under one hour.

Fault conditions in Phase B are simulated by applying controlled values of the Fault Deviation Gain, which emulates the degraded path-holding performance expected during physical wing-loss or rudder-loss events. For each test case, the corresponding .nc file, rudder and wing configuration, and deviation gain are processed through the Digital Twin and Diagnoser to generate predicted vehicle behavior, diagnostic metrics, and a final classification. Across all test cases, the resulting data products, including classification results and runtime measurements, are archived and used to validate compliance with SR-1, SR-2, and SR-4. This System Test Plan is executed in conjunction with the D.A.P. System Test Procedure, which provides the detailed step-by-step instructions for conducting Phase B.

Test Constraints

Constraint	Description	Value
<i>Input File Format</i>	All mission data must be provided as Seaglider-standard .nc files collected during Phase	.nc mission file format and .log

	A.	mission file format
<i>MATLAB Version</i>	All Digital Twin and Diagnoser evaluations must be executed using the same MATLAB environment.	MATLAB R2024b

Test Variables

Variable	Description	Values / Range
<i>Top Rudder</i>	Percentage of rudder mounted on the Seaglider for a given dive	100%, 0%
<i>Port Wing</i>	Percentage of wing mounted on the Seaglider for a given dive	100%, 25%
<i>Fault Deviation Gain</i>	Scaling factor applied to the mission trajectory to mimic degraded path-holding performance when the controller is fighting a real physical fault.	0%, 1%, 2%

Measurements

Measurement	Purpose	Description	Range
<i>Primary Diagnosis</i>	Identify which fault the system classified	Classification result: Nominal, Wing Fault, Rudder Fault, including severity (e.g., 50% rudder, 25% wing).	Nominal, 25% Wing, 50% Wing, 50% Rudder
<i>Runtime</i>	Verify SR-4 computational performance	Wall-clock time required for the DAP system to process the dive from start to classification.	0-60 minutes or 0-3600 seconds

Test Needs

Fault Deviation		Port Wing			
Gain = 0%					
Top Rudder		0%	25%	50%	100%
	0%				
	25%				
	50%				x
	100%			x	x

Fault Deviation		Port Wing			
Gain = 1%					
Top Rudder		0%	25%	50%	100%
	0%				
	25%				
	50%				x
	100%			x	

Fault Deviation		Port Wing			
Gain = 2%		0%	25%	50%	100%
Top Rudder	0%				
	25%				
	50%				x
	100%			x	

Test Matrix

Run	Port Wing	Rudder	Fault Deviation Gain	Dives Folder Name	.nc and .log File Name	Primary Diagnosis	Runtime (sec)
1	100%	100%	0%	Colvos	p1940003	50% Rudder Fault	110.87
2	100%	100%	0%	Colvos	p1940004	25% Wing Fault	131.54
3	100%	100%	0%	Colvos	p1940005	Nominal Case	166.92
4	100%	100%	0%	Colvos	p1940006	25% Wing Fault	108.63
5	100%	100%	0%	Colvos	p1940007	50% Rudder Fault	78.82
6	100%	100%	0%	Edmonds	p1940005	Nominal Case	97.55
7	100%	100%	0%	Edmonds	p1940006	Nominal Case	75.25
8	100%	100%	0%	Edmonds	p1940013	Nominal Case	53.66
9	100%	50%	0%	Edmonds	p1940008	25% Wing Fault	108.55
10	100%	50%	0%	Edmonds	p1940009	Nominal Case	137.12
11	100%	50%	1%	Edmonds	p1940008	25% Wing Fault	100.18
12	100%	50%	1%	Edmonds	p1940009	Nominal Case	386.39

13	100%	50%	2%	Edmonds	p1940008	25% Wing Fault	219.74
14	100%	100%	0%	Edmonds	p1940005	Nominal Case	97.55
15	100%	100%	0%	Edmonds	p1940006	Nominal Case	75.25
16	100%	100%	0%	Edmonds	p1940013	Nominal Case	53.66
17	50%	100%	0%	Edmonds	p1940011	Nominal Case	144.85
18	50%	100%	0%	Edmonds	p1940012	25% Wing Fault	95.79
19	50%	100%	1%	Edmonds	p1940011	Nominal Case	121.09
20	50%	100%	1%	Edmonds	p1940012	25% Wing Fault	246.37
21	50%	100%	2%	Edmonds	p1940011	Nominal Case	120.84
22	100%	50%	0%	Motive	p1950699	Nominal Case	1485.74
23	100%	50%	0%	Motive	p1950700	25% Wing Fault	1945.83
24	100%	50%	0%	Motive	p1950701	50% Rudder Fault	1607.65
25	100%	50%	0%	Motive	p1950702	50% Wing Fault	1792.83
26	100%	50%	0%	Motive	p1950703	50% Rudder Fault	1889.28